Topic 9: From Controlled to Natural Settings (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

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01 USABILITY TESTING

02 CONDUCTING EXPERIMENTS

03 IN-THE-WILD STUDIES

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USABILITY TESTING

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What is Usability Testing?

- Involves testing how people perform on products in controlled settings
- The people being tested are observed and the time it takes them to complete a task and the number and kinds of errors they make are recorded
- Data is recorded on video and key presses are logged
- The data is used to calculate performance times and to identify and explain errors
- User satisfaction is evaluated using questionnaires and interviews
- Observations about how the product is used in more natural contexts, including in-the wild may be included

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Quantitative Measures

- Number of participants successfully completing the task
- Time to complete the task
- Time to complete the task after time away from it
- Number and type of errors per task
- Number of errors per unit of time
- Number of times people navigate to an item such as online help
- Number of people making the same or similar errors

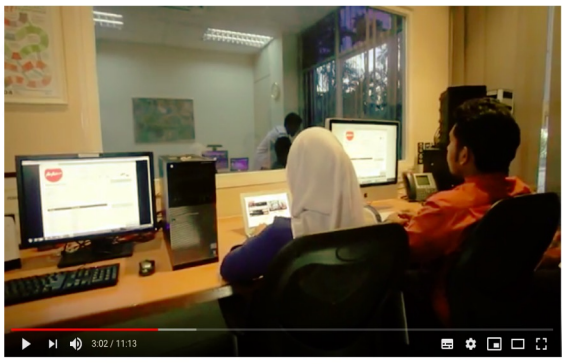
Source: Wixon and Wilson, 1997

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Usability lab with designers watching a user and assistant through a one-way mirror and recording



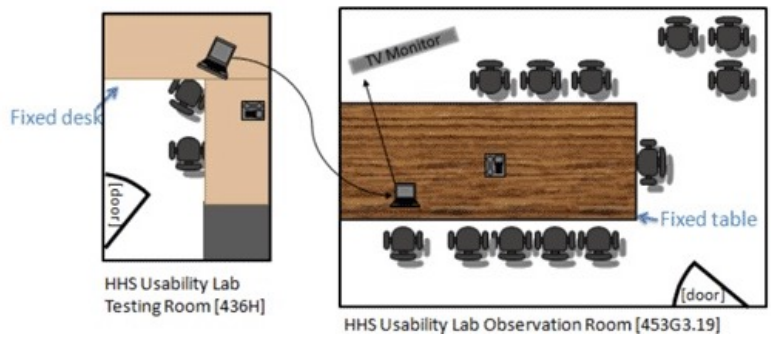


MIMOS UX Lab

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Layout of a usability lab used by US Health & Human Services Dept.



source: https://www.usability.gov/how-to-and-tools/guidance/hhs-usability-lab.html

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Tobii Glasses Mobile Eye-Tracking System



Source: https://www.tobiipro.com/news-events/on-demand-webinars/Wearable-eye-tracking-for-research/



Source: Dalton et al., 2015, p.3891. Reproduced with permission of ACM Publications.

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Equipment is Getting Smaller

- As more testing is done remotely and in-the- wild, equipment has got smaller
- New lightweight models of eye-tracking glasses that look like ordinary glasses
- Video and audio recording is often done using mobile phones which produce increasingly good quality with each new model of phone
- Replace bulky recording devices and tripods, also more discrete

Remote Testing

- Participants can carry out tasks in their own environment without an evaluator being present
- Zoom, Teams and other digital communications are often used
- Testing can be synchronous or asynchronous
- Advantage of remote testing, especially asynchronous testing, is that several participants can be tested at the same time in their own environments
- Covid-19 pandemic encouraged creative ways of conducting remote testing that was safe
- Remote testing may make it easier for people with disabilities to participate
- Downside: no one present to help with equipment problems and less personal than co-present testing

Case 1: iPad usability
(Budiu & Nielsen, 2010)

- Now a classic study – innovative in its day!
- Study conducted in two cities: Fremont, CA and Chicago, IL
- Tests had to be done quickly, as information was needed by third-party app developers
- Also needed to be done secretly so that the competition was not aware of the study before the iPad was launched
- Seven participants with over three months experience with iPhones

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iPad Usability Testing Procedure

- Signed an informed consent form explaining:
  - What the participant would be asked to do
  - The length of time needed for the study
  - The compensation that would be offered for participating
  - Participants' right to withdraw from the study at any time
  - A promise that the person's identity would not be disclosed
  - An agreement that the data collected would be confidential and available only to the evaluators
- Participants were asked to explore the iPad
- Next, they were asked to perform randomly-assigned specified tasks

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Examples of the Tasks used in the iPad Evaluation

App or website	Task
iBook	Download a free copy of *Alice's Adventures in Wonderland* and read through the first few pages.
Craigslist	Find some free mulch for your garden.
eBay	You want to buy a new iPad on eBay. Find one that you could buy from a reputable seller.
Time Magazine	Browse through the magazine and find the best pictures of the week.
Epicurious	You want to make an apple pie for tonight. Find a recipe and see what you need to buy in order to prepare it.
Kayak	You are planning a trip to Death Valley in May this year. Find a hotel located in the park or close to the park.

Adapted from Budiu and Nielsen, 2010
Source: iPad App and Website Usability Study. Used courtesy of the Neilsen Norman Group.

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Problems and Actions

- Examples of problems detected:
  - Accessing the Web was difficult
  - Lack of affordance and feedback
  - Getting lost in an application
  - Knowing where to tap
- Actions by evaluators:
  - Reported to developers
  - Made available to public on Neilsen Norman Group
- Accessibility for all users is important
- Study did not address how iPad would be used in people's everyday lives
- Another study was done a year later to examine this and other issues that there was insufficient time to address in this study

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Case 2: Remote testing with extended VR (XR) (Sanni Siltanen et al., 2021)

- Study at Krone Corp. (Finland) to examine new immersive experiences in VR
- Needed to find ways to test experts and company employees remotely because of Covid-19 pandemic
- Also needed a sophisticated setup to test the collaborative XR platforms across multiple locations
- Testing was conducted remotely with experts in 8 countries: Finland, India, China, Germany, Indonesia, Malaysia, USA, United Arab Emirates
- Some tests in participants premises where special disinfecting routines had to be carried out, other tests were done remotely
- Little advice available about how to do remote testing during a pandemic

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A Participant in a Test Session



Source: Siltanen et al. (2021) MDPI / CC BY 4.0

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Screenshots from the DesignSpace VR Environments

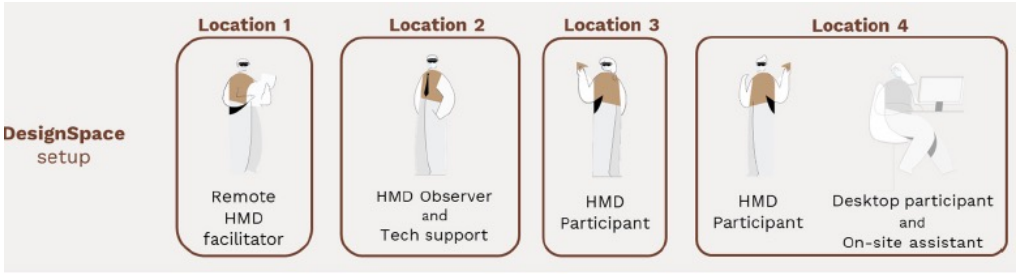


Source: Siltanen et al. (2021) MDPI / CC BY 4.0

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Testing setup used with DesignSpace environments



Source: Siltanen et al. (2021) MDPI / CC BY 4.0

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CONDUCTING EXPERIMENTS

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Experiments

- Test hypotheses
- Predict the relationship between two or more variables
- Independent variable is manipulated by the researcher
- Dependent variable influenced by the independent variable
- Typical experimental designs have one or two independent variables
- Validated statistically and replicable

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Experimental Designs

- Different participants (between participants):
  - Single group of participants is allocated randomly to the experimental conditions
- Same participants (within participants):
  - All participants participate in both conditions
- Matched participants (pairwise design):
  - Participants are matched in pairs, for example, based on expertise, gender, and so on

Different, Same, Matched Participant Design

Design	Advantages	Disadvantages
Different	No order effects	Many subjects and individual differences a problem
Same	Few individuals, no individual differences	Counter-balancing needed because of ordering effects
Matched	Same as different participants, but individual differences reduced	Cannot be sure of perfect matching on all differences

IN-THE-WILD STUDIES

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In-the-Wild Studies

- Done in natural settings
- "In-the-wild" is a term for prototypes being used freely in natural settings – broader and typically less controlled than traditional field studies
- Seek to understand what users do naturally and how technology impacts them
- In-the-wild studies are used in product design to:
  - Identify opportunities for new technology
  - Determine design requirements
  - Decide how best to introduce new technology
  - Evaluate technology in use

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An In-the-wild Study of a Pain-monitoring Device

- Monitoring patients' pain is a known challenge for physicians
- Goal of the study was to evaluate the use of a pain-monitoring device for use after ambulatory surgery
- Painpad is a keypad device
- It was usability tested extensively in the lab before being brought into two hospitals
- Goal was to understand how Painpad was used in the natural environment and as part of routines in two UK hospitals.
- How pain-monitoring changed with Painpad

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Painpad



A tangible device for inpatient self-logging of pain

Source: Price et al., 2018. Reproduced with permission of ACM Publications.

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Data Collection and Participants

- Two studies in two hospitals involving 54 people
- 13 males, 41 females
- Privacy was an important concern
- Hospital stay ranged from 1-7 days, mean and median age 64.6, 64.5 years
- Patients given Painpad after surgery and prompted to report pain levels every two hours
- Nurses also collected scores
- All data entered into charts
- Patients in one hospital were given a user-satisfaction survey when they left
- Also rated Painpad on a 1-5 Likert scale

Data Analysis and Presentation

- Three types of data were collected:
  - Satisfaction with Painpad was based on questionnaire responses
  - Patients' compliance with the two-hour routine
  - How data collected from Painpad compared with data collected by nurses

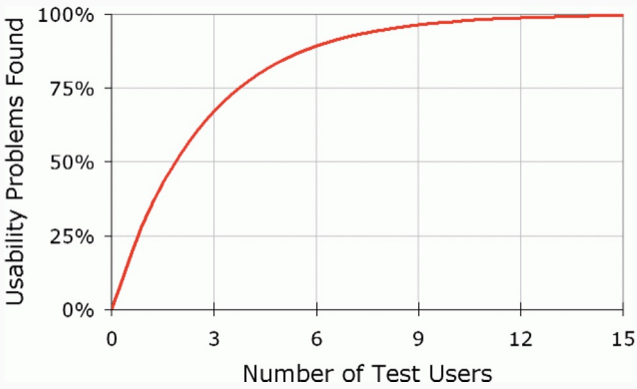
- Data showed:
  - Satisfaction with Painpad 4.63 on Likert scale
  - Patients' compliance was mixed: some liked it while others disliked or didn't notice the prompts
  - Patients recorded more scores with Painpad than through the nurses

How many participants is enough for user testing?

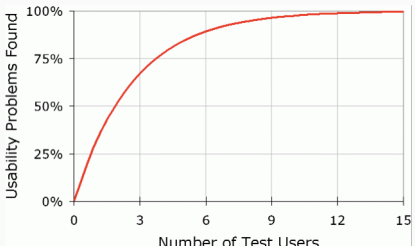
- The number is a practical issue
- Depends on:
  - Schedule for testing
  - Availability of participants
  - Cost of running tests
- Typically 5-10 participants if possible
- Some experts argue that testing should continue until no new insights are gained
- Others suggest that 5 participants is enough to identify any serious problems

Usability testing: Why Need to Test with 5 Users?

The typical value of L is 31%, averaged across a large number of projects studied. Plotting the curve for L =31% gives the following result:



Usability testing: Why Need to Test with 5 Users?



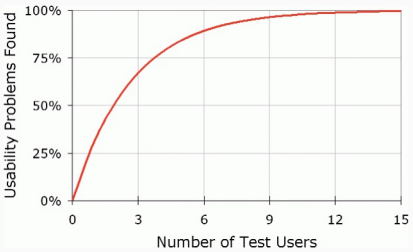
Justify Why 5 users is enough ?

1. As soon as you collect data from a single test user, your insights shoot up and you have already learned almost a third of all there is to know about the usability of the design.

2. When you test the second user, you will discover that this person does some of the same things as the first user, so there is some overlap in what you learn. People are definitely different, so there will also be something new that the second user does that you did not observe with the first user. So the second user adds some amount of new insight, but not nearly as much as the first user did.

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Usability testing: Why Need to Test with 5 Users?



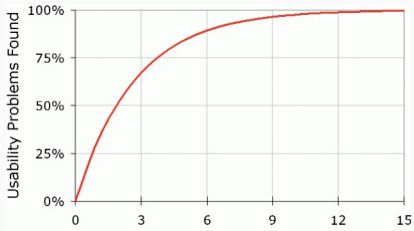
Justify Why 5 users is enough ?

3. The third user will do many things that you already observed with the first user or with the second user and even some things that you have already seen twice. Plus, of course, the third user will generate a small amount of new data, even if not as much as the first and the second user did.

4. As you add more and more users, you learn less and less because you will keep seeing the same things again and again. There is no real need to keep observing the same thing multiple times, and you will be very motivated to go back to the drawing board and redesign the site to eliminate the usability problems.

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Usability testing: Why Need to Test with 5 Users?

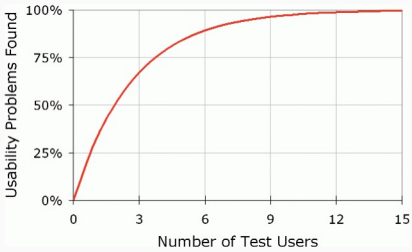


Justify Why 5 users is enough ?

5. After the fifth user, you are wasting your time by observing the same findings repeatedly but not learning much new.

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Usability testing: Why Need to Test with 5 Users?



When you need more users for usability testing ?

When To Test More Users?

You need to test additional users when a website has several **highly distinct groups of users**. The formula only holds for comparable users who will be using the site in fairly similar ways.

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How many participants is enough for other types of evaluation?

- Experiments test hypotheses to discover new knowledge by investigating the relationship between two or more variables
- The number of participants needed depends on the type of experiment being conducted
- It is advisable to consult a statistician before deciding
- In-the-wild studies also vary from a few people in a home to a software team to a whole community

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Usability Testing vs Research

Usability Testing	Experiments for Research
• Improve products	• Discover knowledge
• Few participants	• Often many participants
• Results inform design	• Results validated statistically
• Usually not completely replicable	• Must be replicable
• Conditions controlled as much as possible	• Strongly controlled conditions
• Procedure planned	• Experimental design
• Results reported to developers	• Scientific report to scientific community

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Summary

- Usability testing takes place in controlled spaces - usability labs, temporary labs
- Usability testing focuses on performance measures - how long and how many errors are made when completing a set of predefined tasks
- Indirect observation (video and keystroke logging), user satisfaction questionnaires, and interviews are also collected
- Remote testing has been conducted since the early 1990s but it became important during the Covid19 pandemic
- Remote testing uses portable equipment - video and audio recording using smart phones, mobile eye-tracking and automated keystroke logging

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Summary (cont.)

- Long-term studies of several weeks are used to evaluate complex products that participants need time to learn and use in their own work
- Experiments test a hypothesis by manipulating certain variables while keeping others constant
- The experimenter controls independent variable(s) in order to measure changes in the dependent variable(s)
- In-the-wild studies are carried out in natural settings to discover how people interact with technology in the real world
- In-the-wild studies involve deployment of prototypes or technologies in natural settings
- Sometimes the findings of in-the-wild studies are unexpected, especially for studies that explore how novel technologies are used by participants in their own homes, places of work, or outside

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